Continuous Articulatory Modeling and Causal Control in `covtl-pipeline`

The Nature of the λ Parameter, CC-Branch Extension, and C_2 Invariance in C_1C_2 Sequences

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Abstract

This report formalizes the internal mechanics of the `covtl-pipeline` articulatory synthesis module (driving *VocalTractLab*) as applied to the study of representations in vocal neural networks (notably Wav2Vec2). It rigorously addresses two fundamental questions: (1) **The nature of the λ parameter**: λ is not a simple weighting or a *cross-fade* between two discrete physical states, but corresponds to a continuous vector displacement in the polar control space (ρ, θ) of the consonant branch (CC), coupled with an extension of the organ selector to the union $S_{\text{union}} = S_n \cup S_m$. (2) **The dynamics of C_1C_2 sequences**: In a consonant cluster C_1C_2, the contextual consonant C_2 remains strictly invariant in position (ρ_{C2}, θ_{C2}) and timing, while only the control node of the target consonant C_1 undergoes a continuous causal drift. This asymmetry guarantees perfect causal isolation of the input signal.

1 Introduction and Theoretical Framework

The analysis of place-assimilation processing (e.g., *clean pan* pronounced " clea[m] pan ") by automatic speech recognition (ASR) models rests on the confrontation between natural human speech and synthesized speech. Whereas natural human speech carries confounded variables (coarticulatory micro-cues, prosody, word-frequency biases), articulatory synthesis via `covtl-pipeline` offers a **direct instrument of causal intervention**.

To model the gradual transition from a canonical alveolar consonant /n/ to an assimilated bilabial [m], the experiments rely on a continuum controlled by a parameter $\lambda \in [0, 1]$. The purpose of this document is to formalize the physical and computational reality underlying this parameter λ, and to analyze the behavior of consonant sequences C_1C_2.

2 The Nature of the λ Parameter: Polar Space (ρ, θ) and Selector Union

2.1 Beyond the two-state weighting model

A naive conception of assimilation synthesis would treat λ as a simple acoustic "cross-fade" or a weighted mixture of two pre-existing audio signals. This interpretation is inaccurate. The `covtl-pipeline` model performs no signal mixing; instead, it carries out a **dynamic reallocation within the motor control space of the vocal tract**.

2.2 Representation in the CC-branch polar plane

In the `covtl` articulatory control model, the consonant branch (CC) of a **tautosyllabic cluster** defines each consonant target as a complex position $z_c = \rho e^{i\theta}$ in a polar coordination plane. For the reference pair *clean pan* written as the cluster `klinpan` (k-l-i-n-p-a-n; the dotless form — a syllable dot is a C.C split, a different gesture structure, out of scope of the λ variation):

- **Canonical target /n/ (nasal coda of the cluster)**: $z_c^{[n]} = (\rho = 1.15, \theta = 270°)$ — the engine's cluster rule gives the cluster nasal the stop-like alveolar target.
- **Context target $C_2 = /p/$ (onset of the cluster, fixed)**: $z_c^{[p]} = (\rho = 1.06, \theta = 50°)$.

The parameter λ moves the θ **coordinate of the C_1 target toward the θ of the C_2 target** along the shortest angular path ($\Delta\theta = +140°$ here), with the ρ channel optional:

$$\theta_{C1}(\lambda) = \theta_{C1} + \lambda \, \mathrm{wrap}_{(-\pi,\pi]}(\theta_{C2} - \theta_{C1}), \qquad \rho_{C1}(\lambda) = (1-\lambda)\,\rho_{C1} + \lambda\,\rho_{C2} \text{ (optional)} \tag{1}$$

At $\lambda = 0$ the cluster target is strictly C_1C_2; at $\lambda = 1$ the C_1 polar target has reached the C_2 target coordinates (θ exactly; ρ too when enabled) — $C_1'C_2$: the C_1 gesture now *points at* the C_2 target, while C_2 itself never moves. At $\lambda = 0.5$ with the default (θ-only) channel, the control node sits at ($\rho = 1.15$, $\theta = 340°$), mid-way along the angular path.

2.3 The cluster selector union (S_{union})

The cluster machinery of the engine drives, on every sub-arc of the cluster, the **union of the consonant selectors**:

- Lingual selector of /n/: $S_n = \{\text{JA, TCX, TCY, TBX, TBY, TTX, TTY}\}$.
- Labial selector of /p/: $S_p = \{\text{JA, LD, LP}\}$.
- **Cluster selector**: $S_{\text{union}} = S_n \cup S_p = \{\text{JA, TCX, TCY, TBX, TBY, TTX, TTY, LD, LP}\}$.

This machinery is part of the engine and is *independent of* λ: the variation only moves the target coordinates of C_1 inside it (C_1 keeps its nasal identity — selector and nasality envelope — while its target rotates toward the C_2 place). For every value of λ the trajectory is rebuilt and the full set of 19 anatomical parameters of the vocal tract is recomputed frame by frame (at a rate of 400 Hz).

Summary: What λ physically represents

The λ parameter rotates the polar target of the nasal C_1 toward the place of C_2 inside the fixed cluster machinery (S_{union}). At $\lambda = 0.5$ (reference pair `klinpan`), the system does not superimpose two sounds; it synthesizes a single hybrid gesture — a nasalized approximant whose target sits mid-way along the angular path ($\theta = 340°$): the tongue body is already rotating labial-ward (measured at the C1-window midpoint: jaw $-2.29°$ against $-2.32°$ canonical, tongue-tip x 4.23 cm against 4.78 cm) while the velum stays wide open.

2.4 Levels of description: polar coordination model vs. tract-parameter realization

The polar (ρ, θ) formulation is the *ground truth* of the λ variation (v1.2.0 of covtl-lambda-patch): `vtl-synth morph -targets <cluster>` interpolates the C_1 polar target and re-synthesizes through the engine's own syllabic builder, so every continuum point is a genuine re-synthesized cluster. The older *motor-level* realization — frame-wise linear interpolation of the 19 tract parameters, $X_\lambda = (1-\lambda)A + \lambda B$ between two endpoint sequences — remains available as the blend

option; it approximates the target-level definition and was the only mode of the original kit experiment (which synthesized *dotted* C.C forms and blended them; its recorded mid-window articulator values LP 0.214 → 0.742, LD 0.600 → 0.113, JA −2.638 → −3.309, VO 0.848 belong to that legacy path). Under the cluster target semantic the measured reference values are: JA −2.321 → −3.327, VO 0.886 constant, LP/LD carried by the vowel background at the mid-arc probe (tongue-tip and jaw carry the place change; cf. the patch manual §6).

3 C_1C_2 Sequences: C_2 Invariance and Causal Drift of C_1

3.1 Formalizing consonant pairs

When studying a consonant sequence C_1C_2 in a regressive assimilation context (*clean pan*, where $C_1 = /\text{n}/$ and $C_2 = /\text{p}/$, or *clean tan*, where $C_1 = /\text{n}/$ and $C_2 = /\text{t}/$, realized on the engine as the tautosyllabic clusters *klinpan* / *klintan*), the control trajectory of the CC branch is constructed according to a strict asymmetry.

3.2 Invariance of the contextual consonant C_2

The consonant C_2 serves as the phonological trigger or control context. In the `covtl` pipeline, **the geometric position (ρ_{C2}, θ_{C2}) and the timing of C_2 are strictly invariant** (verified bitwise at every λ by the `outside_c1_arcs_bitwise_canon` check):

- For *clean pan* ($C_2 = /\text{p}/$): Bilabial target fixed at ($\rho = 1.06$, $\theta = 50°$).
- For *clean tan* ($C_2 = /\text{t}/$): Alveolar target fixed at ($\rho = 1.20$, $\theta = -75°$) (engine cluster rule for `klintan`).

3.3 Exclusive causal drift of C_1

Only the target of the nasal consonant C_1 varies as a function of λ:

$$\theta_{C1}^{(\lambda)} = \theta_{[n]} + \lambda \, \text{wrap}\big(\theta_{C2} - \theta_{[n]}\big) \tag{2}$$

Table 1: Evolution of C_1 and C_2 parameters along the λ continuum (reference cluster `klinpan`; θ-only channel, the default)

λ regime	C_1 target in (ρ, θ)	C_2 position (*pan*)	Nature of the C_1 gesture	Assimilation reg
$\lambda = 0.00$	(1.15, 270°)	(1.06, 50°)	Stop-like alveolar nasal	Canonical form /
$\lambda = 0.50$	(1.15, 340°)	(1.06, 50°)	Nasalized approximant	Hybrid / ambiguous
$\lambda = 0.75$	(1.15, 15°)	(1.06, 50°)	Near-labial nasal	Partial assimilat
$\lambda = 1.00$	(1.15, 50°)	(1.06, 50°)	Labial-pointing nasal closure	Target sharing (θ; ρ too with

3.4 Consequences for experimental control

This asymmetry guarantees that the physical modification applied to the C_1 segment is **strictly identical at both the acoustic and articulatory levels**, whether the following consonant C_2 is a /p/ (viable context) or a /t/ (non-viable context).

Major Causal Implication

By construction, the audio signal modifying C_1 carries no information about the context C_2. If the neural network (Wav2Vec2) were to react differently between viable and non-

viable contexts on this purely synthetic signal, it would unambiguously demonstrate the existence of a context-sensitive phonological licensing rule within the network.

4 Implications for Cognitive Modeling and Neural Networks

Simulations conducted on this synthetic continuum for several values of λ ($\lambda \in \{1.0, 0.75, 0.625\}$) yield a fundamental result for the interpretation of the Wav2Vec2 model:

1. **Absence of a viability asymmetry**: At every value of λ, the behavioral compensation rate and the internal probability of the underlying form $P(n)$ remain **strictly equivalent** in the viable context (*clean pan*) and in the non-viable context (*clean tan*).

2. **Demonstrating the nature of the encoder**: The Wav2Vec2 encoder (CNN layers through T6) acts as a **bottom-up acoustic detector**. It possesses no autonomous abstract phonological grammar.

3. **Explaining the effect in human speech**: The asymmetric compensation measured on human speech in the study by Pouw et al. (2024) stems solely from the presence of **physical coarticulatory micro-cues** in the human voice and from the **orthographic statistics of the CTC decoder** (layers T10 through T12).

5 Conclusion

The formalization of the `covtl-pipeline` module demonstrates the high degree of experimental rigor offered by articulatory synthesis: 1. The λ parameter is a **continuous rotation of the C_1 polar target toward the C_2 target** in the polar space (ρ, θ) of the tautosyllabic cluster (θ always, ρ optionally), inside the engine's fixed cluster machinery (selector union S_{union}); at $\lambda = 1$ the C_1 target equals the C_2 target coordinates. Each λ is a genuine re-synthesized cluster (the frame-wise tract-parameter blend of the legacy kit experiment remains available as its motor-level approximation). 2. In a C_1C_2 sequence, the manipulation is **strictly localized on C_1**, keeping C_2 and the whole context bitwise invariant.

This experimental framework makes it possible to definitively dissociate direct acoustic sensitivity from supervised orthographic correction in speech-processing architectures.